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Inventor(s)	Rodeheaver; Austin Xavier et al.

Hybrid power delivery for surgical implants

Abstract

A hybrid power system for delivering an implant to an eye using hydraulic fluid flow or pressure. An implant may be stored, advanced, and delivered to an eye using hydraulic fluid stored in a sterile container through a hollow advancement plunger. The plunger may rigidly advance the implant to a sealed position in a first phase, and then the implant may be advanced into the eye via hydraulic pressure or fluid flow in a second phase. The power system may provide a first cell having a first power density and a second cell having a second power density, wherein the second power density is greater than the first power density. A controller may use the first cell during the first delivery phase and the second cell during the second delivery phase.

Inventors: Rodeheaver; Austin Xavier (Arlington, TX), Taber; Todd (Keller, TX), Van Den Bergh; Roderick (Cambridge, GB), Proulx; Marshall Keith (Keller, TX), Corthorn; Grant (Cambridge, GB), Hemmingway; Christopher (Cambridge, GB), Orrell; Martin (Cambridge, GB), Wyman; Catherine (Cambridge, GB), Penhallurick; Trevor (Cambridge, GB)

Applicant: Alcon Inc. (Fribourg, CH)

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Primary Examiner: Ho; Tan-Uyen T

Assistant Examiner: Igboko; Chima U

Background/Summary

PRIORITY CLAIM (1) This application claims the benefit of priority of U.S. Provisional Patent Application Ser. No. 63/129,026 titled “HYBRID POWER DELIVERY FOR SURGICAL IMPLANTS,” filed on Dec. 22, 2020, whose inventors are Austin Xavier Rodeheaver, Todd Taber, Roderick Van Der Bergh, Marshall Keith Proulx, Grant Corthorn, Chris Hemmingway and Martin Orrell, which is hereby incorporated by reference in its entirety as though fully and completely set forth herein.

TECHNICAL FIELD

(1) The invention set forth in the appended claims relates generally to eye surgery. More particularly, but without limitation, the claimed subject matter relates to systems, apparatuses, and methods for inserting an implant into an eye.

BACKGROUND

(2) The human eye can suffer a number of maladies causing mild deterioration to complete loss of vision. While contact lenses and eyeglasses can compensate for some ailments, ophthalmic surgery may be required for others. In some instances, implants may be beneficial or desirable. For example, an intraocular lens may replace a clouded natural lens within an eye to improve vision.

(3) While the benefits of intraocular lenses and other implants are known, improvements to delivery systems, components, and processes continue to improve outcomes and benefit patients.

BRIEF SUMMARY

(4) New and useful systems, apparatuses, and methods for eye surgery are set forth in the appended claims. Illustrative embodiments are also provided to enable a person skilled in the art to make and use the claimed subject matter.

(5) For example, some embodiments may comprise or consist essentially of an apparatus for delivering an implant, such as an intraocular lens. In more particular examples, the apparatus may comprise a rigid plunger for advancing an implant. Some embodiments may additionally comprise a bore through the rigid plunger, which can allow a working fluid to advance the implant into the eye via hydraulic pressure. For example, a hollow rigid plunger can be used to first advance an intraocular lens to a point that a seal is created about the intraocular lens within a delivery lumen. The lens may then be hydraulically advanced to delivery by passing a working fluid through the hollow bore of the plunger.

(6) Some embodiments may comprise or consist essentially of a hybrid power module having a primary cell and a secondary cell. In some embodiments, for example, the primary cell and the secondary cell may be a primary battery and a secondary battery, wherein the primary battery and the secondary battery have different power densities. In other examples, the primary cell may be a battery and the secondary cell may be a capacitor. Each cell can provide distinct power delivery capabilities at different times during the delivery procedure. For example, the primary cell may provide a relatively lower power for initial movement of an implant over a relatively longer range in the first phase, and the secondary cell may provide a relatively higher peak power for delivery of the implant through the delivery lumen in the second phase.

(7) More generally, some embodiments may comprise or consist essentially of an apparatus for operating an implant delivery device. Such embodiments may comprise a motor, a primary cell, a secondary cell, and a controller. The motor may be configured to be coupled to the implant delivery device. The primary cell may have a first energy density and a first power density; and the secondary cell may have a second energy density and a second power density. The controller may be coupled to the motor, the primary cell, and the secondary cell. The controller may be configured to selectively couple the primary cell to the motor for a first delivery range, couple the secondary cell to the motor for a second delivery range, and couple the primary cell to the secondary cell for a charging period. In more particular embodiments, the second power density may be greater than the first power density.

(8) Some embodiments of an apparatus for delivering an implant to an eye may comprise a nozzle having a delivery lumen, an implant bay coupled to the nozzle, an actuator, a motor configured to be coupled to the actuator, a primary cell, and a secondary cell. A controller may be coupled to the motor, the primary cell, and the secondary cell. The controller may be configured to selectively couple the primary cell and the secondary cell to the motor. In some embodiments, the controller may couple the primary cell to the motor to operate the actuator to drive the implant from a first position to a second position and may couple the secondary cell to the motor to operate the actuator

to drive the implant to a third position. In more particular embodiments, the actuator may comprise a push rod configured to engage the implant. For example, the push rod may be a rigid plunger in some embodiments. Additionally, some embodiments of the actuator may comprise a bore through the push rod, which can be fluidly coupled to the delivery lumen in the nozzle. For example, the push rod may be a hollow, rigid plunger in some embodiments. The implant may be a lens in some embodiments.

(9) A method for delivering or ejecting an implant from a delivery system may comprise providing the implant in an implant bay, applying a first delivery force to advance the implant from the implant bay to a delivery lumen with a rigid push rod, and applying a second delivery force to advance the implant through the delivery lumen. The second delivery force may be greater than the first delivery force. In some embodiments, the push rod may comprise or consist essentially of a rigid plunger. In yet more particular embodiments, the method may additionally comprise moving a working fluid through a bore in the rigid plunger with the second delivery force.

(10) Features, elements, and aspects described in the context of some embodiments may also be omitted, combined, or replaced by alternative features. Other features, objectives, advantages, and a preferred mode of making and using the claimed subject matter are described in greater detail below with reference to the accompanying drawings of illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings illustrate some objectives, advantages, and a preferred mode of making and using some embodiments of the claimed subject matter. Like reference numbers represent like parts in the examples.

(2) FIG. 1 is a schematic view of an example system for inserting an implant into an eye.

(3) FIG. 2 is a schematic diagram of an example of a delivery module that may be associated with some embodiments of the system of FIG. 1.

(4) FIG. 3 is a detail view of an actuator that may be associated with the example of FIG. 2.

(5) FIG. 4 is a schematic diagram of an example of a drive module that may be associated with some embodiments of the system of FIG. 1.

(6) FIG. 5 is a schematic diagram of an example power module that may be associated with the drive module of FIG. 4.

(7) FIG. 6 is a schematic diagram of an example power profile that may be associated with the example power module of FIG. 5.

(8) FIGS. 7A-7C are schematic diagrams illustrating an example method of ejecting an implant from the system of FIG. 1.

(9) FIG. 8A-8B are schematic diagrams illustrating an example application of the system of FIG. 1 to insert an implant into an eye.

DESCRIPTION OF EXAMPLE EMBODIMENTS

(10) The following description of example embodiments provides information that enables a person skilled in the art to make and use the subject matter set forth in the appended claims, but it may omit certain details already well known in the art. The following detailed description is, therefore, to be taken as illustrative and not limiting.

(11) The example embodiments may also be described herein with reference to spatial relationships between various elements or to the spatial orientation of various elements depicted in the attached drawings. In general, such relationships or orientation assume a frame of reference consistent with or relative to a patient in a position to receive an implant. However, as should be recognized by those skilled in the art, this frame of reference is merely a descriptive expedient rather than a strict prescription.

(12) FIG. 1 is a schematic diagram of a system **100** that can insert an implant into an eye. In some embodiments, the system **100** may comprise two or more modules, which can be configured to be coupled and decoupled as appropriate for storage, assembly, use, and disposal. For example, as illustrated in FIG. 1, some embodiments of the system **100** may include a nozzle **105**, an implant bay **110** coupled to the nozzle **105**, and an actuator **115** coupled to the implant bay **110**. In some embodiments, the system **100** may additionally comprise a drive module **120** configured to engage the actuator **115**.

(13) The nozzle **105** generally comprises a tip adapted for insertion through an incision into an eye. The size of the tip may be adapted to surgical requirements and techniques as needed. For example, small incisions are generally preferable to reduce or minimize healing times. Incisions of less than 3 millimeters may be preferable in some instances, and the tip of the nozzle **105** may have a width of less than 3 millimeters in some embodiments.

(14) The implant bay **110** generally represents a wide variety of apparatuses that are suitable for storing an implant prior to delivery into an eye. In some embodiments, the implant bay **110** may additionally or alternatively be configured to prepare an implant for delivery. For example, some embodiments of the implant bay **110** may be configured to be actuated by a surgeon or other operator to prepare an implant for delivery by subsequent action of the actuator **115**. In some instances, the implant bay **110** may be configured to actively deform, elongate, extend, or otherwise manipulate features of the implant before the implant is advanced into the nozzle **105**. For example, the implant bay **110** may be configured to extend or splay one or more features, such as haptics, of an intraocular lens.

(15) The actuator **115** is generally configured to advance an implant from the implant bay **110** into the nozzle **105**, and thereafter from the nozzle **105** through an incision and into an eye.

(16) The drive module **120** is generally operable to energize the actuator **115**. In some examples, the drive module **120** may be operated by electrical, mechanical, hydraulic, or pneumatic power, or combinations thereof, or in some other manner. In some instances, the drive module **120** may be operated manually. According to other implementations, the drive module **120** may be an automated system.

(17) In general, components of the system **100** may be coupled directly or indirectly. For example, the nozzle **105** may be directly coupled to the implant bay **110** and may be indirectly coupled to the actuator **115** through the implant bay **110**. Coupling may include fluid, mechanical, thermal, electrical, or chemical coupling (such as a chemical bond), or some combination of coupling in some contexts. For example, the actuator **115** may be mechanically coupled to the drive module **120** and may be mechanically and fluidly coupled to the nozzle **105**. In some embodiments, components may also be coupled by virtue of physical proximity, being integral to a single structure, or being formed from the same piece of material.

(18) FIG. 2 is a schematic diagram of an example of a delivery module **200** that may be associated with some embodiments of the system **100**. In the example of FIG. 2, delivery module **200** comprises the nozzle **105**, the implant bay **110**, and the actuator **115**. The nozzle **105** of FIG. 2 has a delivery lumen **205**, and an implant **210** is disposed within the implant bay **110**.

(19) The actuator **115** of FIG. 2 generally comprises a housing **215** and a push rod, such as a plunger **220**, disposed within the housing **215**. In some embodiments, the plunger **220** or other push rod may be comprised of a substantially rigid material, such as a medical grade polymer material. In the example of FIG. 2, the actuator **115** further comprises a bore **225** through the plunger **220** and a driver interface **230**. A plunger seal **235** may be disposed within the housing **215** and coupled to the plunger **220**. A drive seal **240** may also be disposed within the housing **215**.

(20) As illustrated in the example of FIG. 2, the drive seal **240** may be disposed between the plunger seal **235** and the drive interface **230**, and a fluid chamber **250** may be defined within the housing **215** between the plunger seal **235** and the drive seal **240**. In the example configuration of FIG. 2, the plunger seal **235** is configured to provide a fluid seal across the housing **215** and

substantially prevent movement of fluid from the fluid chamber **250** to the bore **225**. The drive seal **240** may also be configured to provide a fluid seal across the housing **215** and substantially prevent movement of fluid from the fluid chamber **250** to the drive interface **230**.

(21) FIG. **3** is a detail view of the actuator **115** of FIG. **2**, illustrating additional details that may be associated with some embodiments. For example, the housing **215** of FIG. **3** further comprises a plunger interface **305** and a bypass channel **310** disposed between the plunger interface **305** and the drive interface **230**. The bypass channel **310** may take various forms. For example, the bypass channel **310** may comprise a protrusion in the housing **215**, as illustrated in FIG. **3**. In other examples, the bypass channel **310** may comprise a groove or recess in the inner surface of the housing **215**. In some embodiments, the bypass channel **310** may comprise a plurality of channels. For example, a plurality of channels may be disposed circumferentially around the housing **215** in some embodiments.

(22) The plunger **220** generally has a first end **315** and a second end **320**, wherein the first end **315** is generally disposed adjacent to the plunger interface **305**. The bore **225** generally passes through the plunger **220** longitudinally from the first end **315** to the second end **320**.

(23) In some embodiments, the actuator **115** may additionally comprise a nozzle seal **325** and a bypass seal **330**. Each of the nozzle seal **325** and the bypass seal **330** are generally configured to create a seal between a portion of the plunger **220** and the housing **215** to substantially prevent movement of fluid past the seal. As illustrated in the example of FIG. **3**, one or both of the nozzle seal **325** and the bypass seal **330** may be ring seals, such as an O-ring, disposed circumferentially around a portion of the plunger **220**. In other examples, an umbrella seal may be suitable. In more particular embodiments, the nozzle seal **325** may be disposed proximate to the first end **315** of the plunger **220**, and the bypass seal **330** may be disposed proximate to the second end **320** of the plunger **220**.

(24) The drive interface **230** of FIG. **3** comprises a cap **335** and an aperture **340**. The cap **335** may be coupled to an end of the housing **215** to retain the drive seal **240** and other components within the housing **215**.

(25) FIG. **4** is a schematic diagram of an example of the drive module **120** of FIG. **1**, illustrating additional details that may be associated with some embodiments. For example, the drive module **120** of FIG. **4** generally comprises a power module **400** and a motor **405** coupled to the power module **400**. A drive shaft **410** may be coupled to the motor **405**. In some embodiments, the drive shaft **410** may comprise or consist essentially of a lead screw. In some embodiments, the drive shaft **410** may be coupled to an actuator interface **415**. The drive module **120** may also comprise a switch **420** for variably controlling the speed of the motor **405**, and an encoder wheel **425** for measuring movement of the drive shaft **410**.

(26) FIG. **5** is a schematic diagram illustrating an example of the power module **400** that may be associated with some examples of the drive module **120**. In the example of FIG. **5**, the power module **400** generally comprises a primary cell **505**, a secondary cell **510**, and a controller **515** coupled to the primary cell **505** and the secondary cell **510**. The primary cell **505** of FIG. **5** may have a first energy density and a first power density, and the secondary cell **510** may have a second energy density and a second power density. In some embodiments, the first energy density may be greater than the second energy density, and the second power density may be greater than the first power density. For example, the primary cell **505** may be a battery, which can provide a relatively higher energy density and lower power density, and the secondary cell **510** may be a capacitor, which can provide a relatively higher power density and lower energy density. In some examples, the primary cell **505** may be a lithium-based battery, and the secondary cell **510** may be a supercapacitor. As illustrated in FIG. **5**, some embodiments may additionally comprise a user interface **520**, one or more power conditioning units **525**, and one or more electronic switching devices, such as solid-state relays **530**.

(27) The controller **515** may be configured to selectively couple the primary cell **505** to the motor

405 for a first delivery range and couple the secondary cell **510** to the motor **405** for a second delivery range. For example, in some embodiments, the controller **515** may control the solid-state relays **530** to couple and disconnect the motor **405** to the primary cell **505** and the secondary cell **510**. In some embodiments, the higher energy density and lower power density of the primary cell **505** can provide a relatively lower power to the motor **405** over a longer duration, and the lower energy density and higher power density of the secondary cell **510** can provide a relatively higher power to the motor **405** over a shorter duration. The controller **515** may also be configured to selectively couple the primary cell **505** to the secondary cell **510** for a charging period so that the primary cell **505** can charge the secondary cell **510**. In some embodiments, the power module **400** may also comprise a switchable load resistor, which can reduce leakage current during storage and autoclave cycles. Additionally, or alternatively, the controller **515** may selectively couple the secondary cell **510** to the resistor to discharge the secondary cell **510**. For example, the secondary cell **510** may be discharged before an autoclave cycle. A thermoelectric generator may also be used to charge the primary cell **505**, the secondary cell **510**, or both during an autoclave cycle.

(28) FIG. **6** is a simplified chart illustrating an example power profile that may be associated with some embodiments of the power module **400** of FIG. **5**. More particularly, the chart of FIG. **6** illustrates a force that can be provided by the motor **405** if selectively coupled to the primary cell **505** and the secondary cell **510** over different delivery ranges.

(29) In some embodiments, the delivery ranges may be based on movement of the drive shaft **410**. For example, the encoder wheel **425** may provide a signal to the controller **515** that is indicative of a position of the drive shaft **410**, such as the distance of the drive shaft **410** from a nominal starting point **D0**. Additionally, or alternatively, the delivery ranges may be based on energy measurements, such as current measured from the motor **405**. For example, a current sensor (not shown) may provide a signal to the controller **515** that is indicative of power requirements from the motor **405**, and the controller **515** may switch between the primary cell **505** and the secondary cell **510** over different delivery ranges based on this signal. In the example of FIG. **6**, the controller **515** couples the primary cell **505** to the motor **405** for a first delivery range of **D0** to **D1**, and the motor **405** can provide a delivery force to the drive shaft **410** in a first force range of **F0** to **F1**. In a second delivery range of **D1** to **D2**, the controller **515** may switch the motor **405** to the secondary cell **510**, and the motor **405** can provide a delivery force to the drive shaft **410** in a second force range of **F1** to **F2**. In a third delivery range of **D2** to **D3**, the controller **515** may switch the motor **405** back to the primary cell **505**, and the motor **405** can provide a delivery force in the first force range of **F0** to **F1**. In the example of FIG. **6**, the second delivery range of **D1** to **D2** is shorter than the first delivery range of **D0** to **D1**, and the first force range of **F0** to **F1** is less than the second force range of **F1** to **F2**.

(30) FIGS. **7A-7C** are schematic diagrams illustrating an example method of ejecting the implant **210** from the system **100**. Initially, various components of the system **100** may be assembled if needed. For example, the nozzle **105**, the implant bay **110**, and the actuator **115** may be coupled to each other as illustrated in FIG. **7A**. The drive module **120** may also be coupled to the actuator **115** through the drive interface **230**. For example, the drive shaft **410** may be configured to directly engage the drive seal **240** through the drive interface **230**, as illustrated in FIG. **7A**. In other examples, the actuator interface **415** may be configured to engage the drive seal **240** through the drive interface **230**. In some embodiments, the drive interface **230** may comprise an aperture configured to receive the drive shaft **410**, the actuator interface **415**, or both.

(31) The implant **210** may be provided in the implant bay **110**, as illustrated in the example of FIG. **7A**. In some embodiments, the implant **210** may comprise an intraocular lens, which may have a shape similar to that of a natural lens of an eye, and it may be made from numerous materials. In the example of FIG. **7A**, the implant **210** is illustrative of an intraocular lens having an optic body **705**, a leading haptic **710**, and a trailing haptic **715**. Examples of suitable materials may include silicone, acrylic, and combinations of such suitable materials. In some instances, the implant **210**

may comprise an intraocular lens that is fluid-filled, such as a fluid-filled accommodating intraocular lens.

(32) The plunger **220**, the plunger seal **235**, and the drive seal **240** are generally movable within the housing between a first position, as illustrated in the example of FIG. 7A, and other positions illustrated in FIG. 7B-7C.

(33) In some examples, a working fluid **720** may be stored in the fluid chamber **250**. In the first position of FIG. 7A, the plunger seal **235** fluidly isolates the bore **225** from the working fluid **720** in the fluid chamber **250**, which can allow the working fluid **720** to be stored within the fluid chamber **250** in the first position. In some examples, the nozzle seal **325** and the first end **315** of the plunger **220** may protrude into the implant bay **110** in the first position, as illustrated in FIG. 7A, which can create a seal in the implant bay **110** behind the implant **210**. The first end **315** of the plunger **220** may also engage the implant **210** in the first position, in some examples. In other examples, the nozzle seal **325** and the first end **315** may be contained within the housing **215** in the first position.

(34) In some embodiments, the drive module **120** may move the drive shaft **410** against the drive seal **240**, which can rigidly move the plunger **220**, the plunger seal **235**, the drive seal **240**, and the working fluid **720**, maintaining a fixed relationship as illustrated in FIG. 7B. For example, the controller **515** may engage the motor **405** to the primary cell **505** to provide a delivery force in the first range of **F0** to **F1** to the drive shaft **410**, and the delivery force of the drive shaft **410** can move the plunger **220**, the plunger seal **235**, the drive seal **240**, and the working fluid **720** from the first position of FIG. 7A to the second position of FIG. 7B. In some embodiments, the movement from the first position to the second position may correlate to moving the drive shaft **410** through the first delivery range of **D0** to **D1** in FIG. 6.

(35) In the position of FIG. 7B, the implant **210** is advanced into the delivery lumen **205**, which may create a fluid seal between the implant **210** and the delivery lumen **205**. In some examples, the implant **210** may be positioned entirely within the delivery lumen **205**. In the second position, the bypass channel **310** fluidly couples the bore **225** to the fluid chamber **250** around the plunger seal **235**. As the drive shaft **410** and the drive seal **240** apply pressure to the working fluid **720** in the fluid chamber **250**, the working fluid **720** may move into the bore **225** through the bypass channel **310**, unimpeded at a higher flow rate.

(36) The plunger **220** may be retained in the second position of FIG. 7B against further force applied to the drive seal **240**. For example, in some embodiments, the second end **320** of the plunger **220** may be flared, and the plunger interface **305** may be configured to engage the second end **320** to limit advancement. Additionally, or alternatively, the implant bay **110** or the nozzle **105** may comprise a plunger stop **725** configured to engage some portion or feature of the plunger **220**, such as the second end **320** of the plunger **220**, to prevent further advancement. In yet other examples, some embodiments of the delivery lumen **205** may be tapered, which can prevent further advancement of the plunger **220** into the delivery lumen **205**. For example, the diameter of the delivery lumen **205** may decrease as it gets further from the implant bay **110**.

(37) With the plunger **220** retained, additional pressure applied by the drive seal **240** on the working fluid **720** can move the working fluid **720** through the bypass channel **310** and the bore **225**, as illustrated in the example of FIG. 7C. Movement of the working fluid **720** from the bore **225** into the delivery lumen **205** under pressure from the drive seal **240** can increase the pressure and flow rate of the working fluid **720** in the delivery lumen **205** behind the implant **210**, which can advance the implant **210** further through the delivery lumen **205** until the implant **210** is ejected. In some embodiments, additional force from the drive shaft **410** may be advantageous for moving the working fluid **720** through the bore **225** and the delivery lumen **205** behind the implant **210** and moving the implant **210** through the delivery lumen **205**. To provide additional force, the controller **515** may switch the motor **405** to the secondary cell **510** to increase the power density available to the motor **405**. For example, the secondary cell **510** may provide a higher power density than the

primary cell **505**, which can be used to provide a delivery force in the second range of F1 to F2 to drive the drive shaft **410** from the second position of FIG. 7B to the third position of FIG. 7C, which may correlate to the second delivery range of D1 to D2 in FIG. 6.

(38) FIGS. 8A-8B are schematic diagrams further illustrating an example use of the system **100** to deliver the implant **210** to an eye **800**. As illustrated, an incision **805** may be made in the eye **800** by a surgeon, for example. In some instances, the incision **805** may be made through the sclera **810** of the eye **800**. In other instances, an incision may be formed in the cornea **815** of the eye **800**. The incision **805** may be sized to permit insertion of a portion of the nozzle **105** in order to deliver the implant **210** into the capsular bag **820**. For example, in some instances, the size of the incision **805** may have a length less than about 3000 microns (3 millimeters). In other instances, the incision **805** may have a length of from about 1000 microns to about 1500 microns, from about 1500 microns to about 2000 microns, from about 2000 microns to about 2500 microns, or from about 2500 microns to about 3000 microns.

(39) After the incision **805** is made, the nozzle **105** can be inserted through the incision **805** into an interior portion **825** of the eye **800**. The system **100** can then eject the implant **210** through the nozzle **105** into the capsular bag **820** of the eye **800**, substantially as described above with reference to FIGS. 7A-7C. In some applications, the implant **210** may be delivered with one or more of the leading haptic **710** and the trailing haptic **715** in a folded configuration and can revert to an initial, unfolded state, within the capsular bag **820**, as shown in FIG. 8B. The capsular bag **820** can retain the implant **210** within the eye **800** in a relationship relative to the eye **800** so that the optic body **705** refracts light directed to the retina (not shown). The leading haptic **710** and the trailing haptic **715** can engage the capsular bag **820** to secure the implant **210** therein. After dispensing the implant **210** into the capsular bag **820**, the nozzle **105** may be removed from the eye **800** through the incision **805**, and the eye **800** can be allowed to heal over a period of time.

(40) The systems, apparatuses, and methods described herein may provide significant advantages. For example, some embodiments may be particularly advantageous for delivering intraocular lenses, including fluid-filled accommodating lenses, which can present unique challenges for delivery. Some embodiments can compress a relatively large lens to fit through an acceptably small incision, manage deformation caused by shifting fluid during compression and exit from a nozzle, and execute delivery in a predictable and controlled manner. Additionally, some embodiments can reduce system complexity and the number of delivery steps while maintaining haptic position consistency. Some embodiments may also reduce the amount of working fluid for delivery.

(41) Additionally, or alternatively, some embodiments may provide a hybrid power delivery, which may be advantageous for delivering some implants. For example, some embodiments of the power module **400** may provide both a relatively lower power over a longer distance and a relatively higher peak power over a shorter distance. In some embodiments of the system **100**, the implant **210** may be larger than the delivery lumen **205** and may benefit from the hybrid power profile provided by some embodiments of the power module **400** that can provide lower power to move the implant **210** into the delivery lumen **205** and increased power to move the implant **210** through the delivery lumen **205**. Hybrid power may also support long-term, low-level energy requirements, which may be advantageous for some embodiments. More particular advantages may include reducing the cost and complexity of batteries and other power cells.

(42) Actuation force experienced by an operator may also be reduced in some embodiments. For example, a surgeon may only feel the actuation force of the switch, which can be significantly lower than some types of mechanical drive systems. Additionally, or alternatively, some embodiments may include facilitating one-handed operation and reversal, which can also reduce the number of staff required to perform a surgical procedure.

(43) While shown in a few illustrative embodiments, a person having ordinary skill in the art will recognize that the systems, apparatuses, and methods described herein are susceptible to various changes and modifications that fall within the scope of the appended claims. Moreover,

descriptions of various alternatives using terms such as “or” do not require mutual exclusivity unless clearly required by the context, and the indefinite articles “a” or “an” do not limit the subject to a single instance unless clearly required by the context. Components may be also be combined or eliminated in various configurations for purposes of sale, manufacture, assembly, or use. For example, in some configurations, the nozzle **105**, the implant bay **110**, the actuator **115**, the drive module **120** may each be separated from one another or combined in various ways for manufacture or sale.

(44) The claims may also encompass additional subject matter not specifically recited in detail. For example, certain features, elements, or aspects may be omitted from the claims if not necessary to distinguish the novel and inventive features from what is already known to a person having ordinary skill in the art. Features, elements, and aspects described in the context of some embodiments may also be omitted, combined, or replaced by alternative features serving the same, equivalent, or similar purpose without departing from the scope of the invention defined by the appended claims.

Claims

1. An apparatus for operating an implant delivery device, the apparatus comprising: a motor; a drive shaft coupled to the motor and configured to be coupled to the implant delivery device, the drive shaft having a first delivery range and a second delivery range; a primary cell having a first energy density and a first power density; a secondary cell having a second energy density and a second power density; and a controller coupled to the motor, the primary cell, and the secondary cell; the controller configured to: receive a signal indicative of a position of the drive shaft; and based on the signal indicating the position is in the second delivery range, perform at least both of: (i) disconnecting the primary cell from the motor, and (ii) coupling the secondary cell to the motor.
 2. The apparatus of claim 1, wherein the second power density is greater than the first power density.
 3. The apparatus of claim 1, wherein the first energy density is greater than the second energy density.
 4. The apparatus of claim 1, wherein the first delivery range is greater than the second delivery range.
 5. The apparatus of claim 4, wherein the controller is further configured to couple the primary cell to the motor if the position is in the first delivery range.
 6. The apparatus of claim 1, wherein at least one of the primary cell and the secondary cell is a battery.
 7. The apparatus of claim 1, wherein the secondary cell is a capacitor.
 8. The apparatus of claim 1, wherein the secondary cell is a supercapacitor.
 9. The apparatus of claim 1, wherein the controller is further configured to selectively couple the primary cell to the secondary cell for a charging period.
 10. The apparatus of claim 1, wherein the controller is further configured to discharge the secondary cell.
 11. The apparatus of claim 1, further comprising a thermoelectric generator configured to charge at least one of the primary cell and the secondary cell.
 12. The apparatus of claim 1, wherein the controller is further configured to couple the primary cell to the motor if the position is in the first delivery range.
 13. The apparatus of claim 1, wherein the controller is configured to couple the primary cell to the motor based on the signal indicating the position is in the first delivery range.
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